EXHAUST GAS RECIRCULATION SYSTEM [SKYACTIV-D 1.5]

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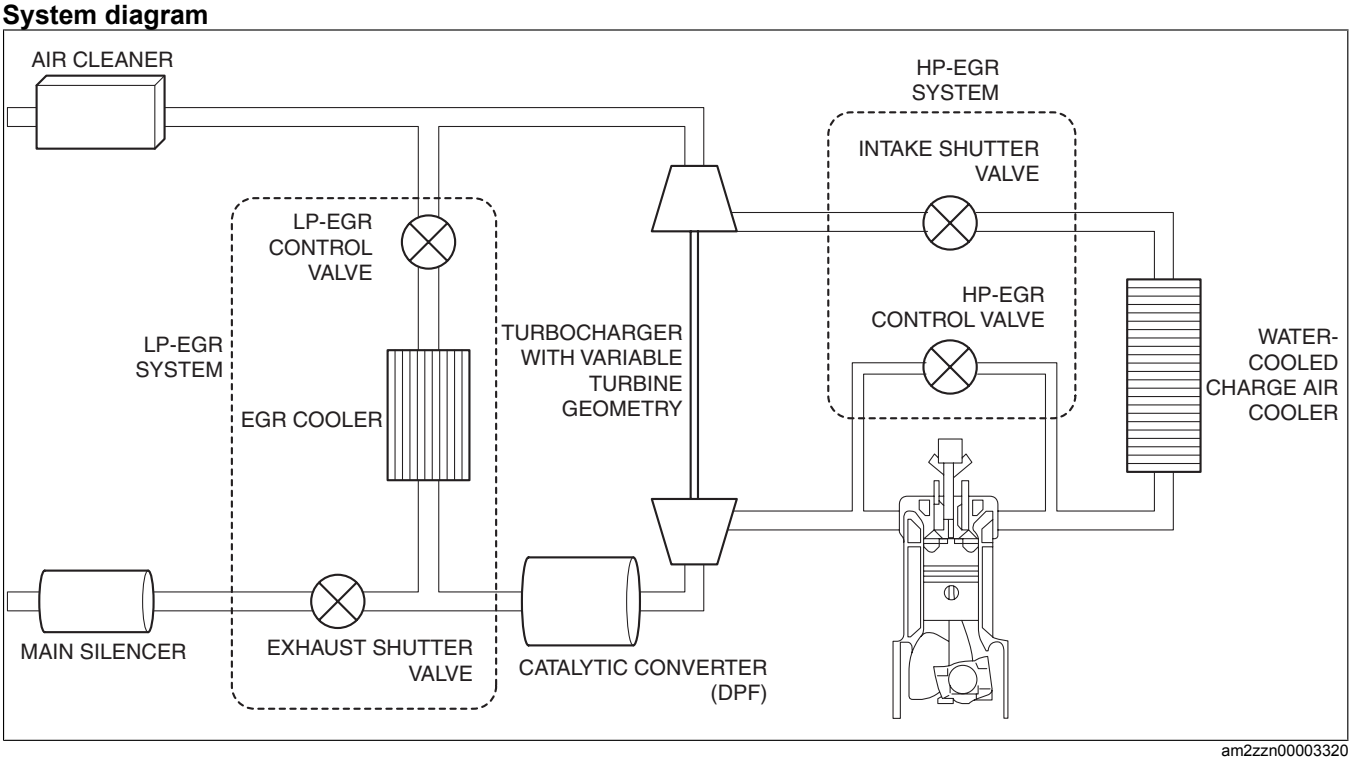
Purpose, Outline

- The catalytic converter recirculates the exhaust gas in the combustion chamber and lowers the combustion temperature. By recirculating the exhaust gas, the NOx generated largely when the combustion temperature is high is reduced.
- The HP-EGR system*1 and LP-EGR system*2 are used concurrently to perform optimum control according to the engine operation conditions.

*1 : The HP-EGR system circulates/recirculates exhaust gas in the intake manifold prior to it passing the turbine of the variable geometry turbocharger. Because the exhaust gas is recirculated immediately, responsiveness to the necessity of nitrous oxide concentration adjustment and the temperature increasing ability during engine warm-up are excellent.

*2 : The LP-EGR system sends/recirculates exhaust gas to the variable geometry turbocharger inlet after it passes the variable geometry turbocharger turbine. Because the intake air and exhaust gas are charged together, even high volumes in the charge region of the exhaust gas can be recirculated.

System diagram



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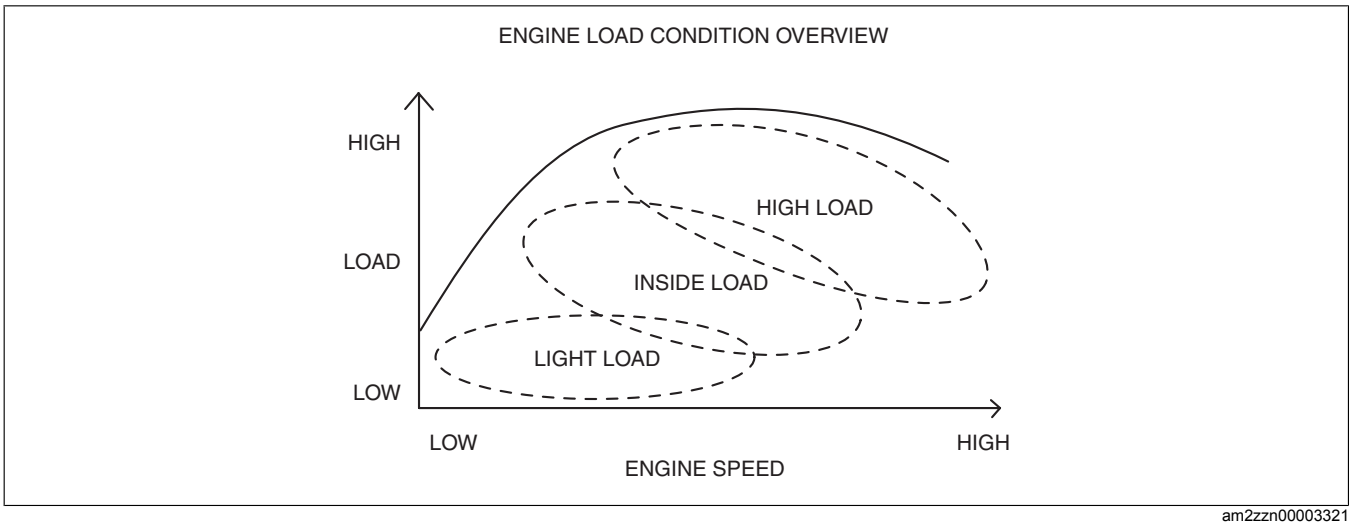
Structure

- The exhaust gas recirculation system consists of the following parts:

Part, system name		Reference
HP-EGR control valve	Intake shutter valve	(See INTAKE SHUTTER VALVE [SKYACTIV-D 1.5].)
	HP-EGR control valve	(See HP-EGR CONTROL VALVE [SKYACTIV-D 1.5].)
LP-EGR control valve	Exhaust shutter valve	(See EXHAUST SHUTTER VALVE [SKYACTIV-D 1.5].)
	LP-EGR control valve	(See LP-EGR CONTROL VALVE [SKYACTIV-D 1.5].)
	EGR cooler	(See EGR COOLER [SKYACTIV-D 1.5].)

Operation

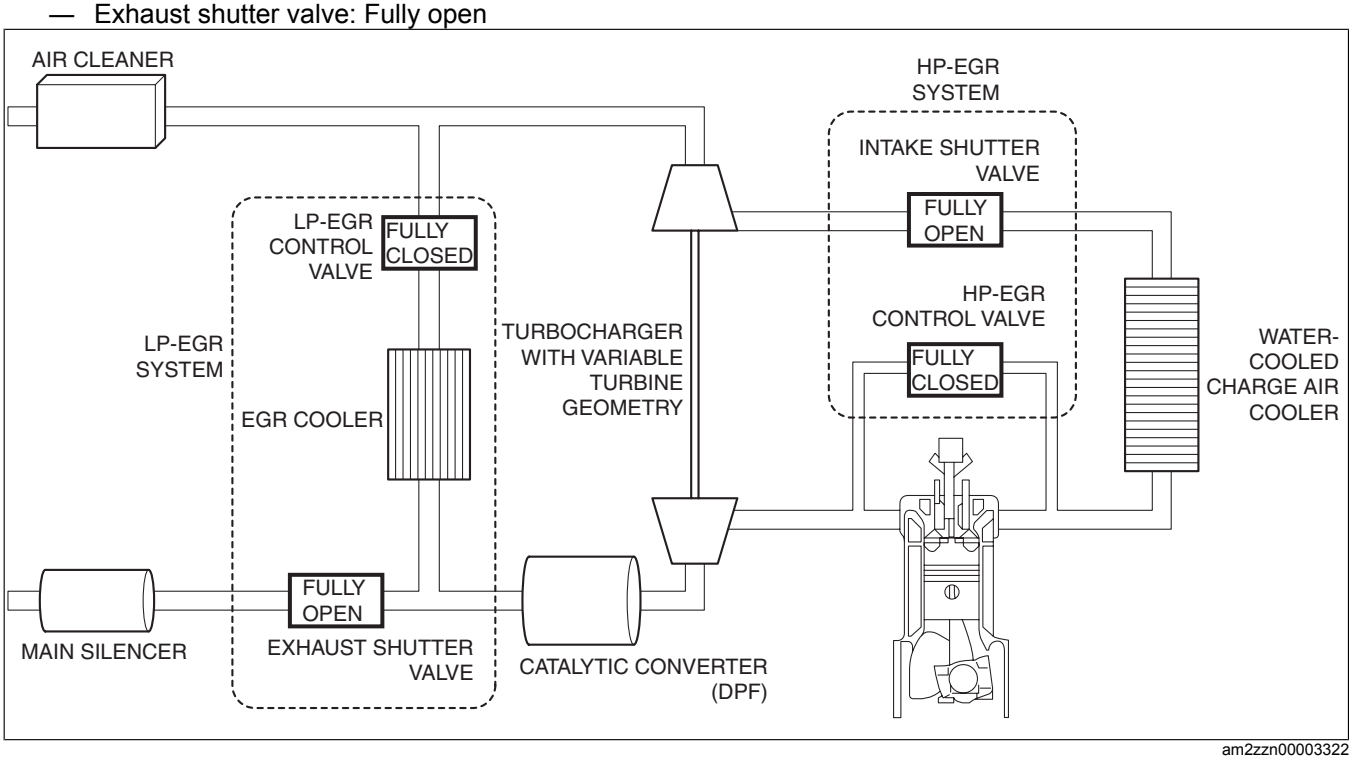
- The exhaust gas recirculation system operates the HP-EGR and LP-EGR systems according to the engine load.



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Ignition switched off (IG-OFF)

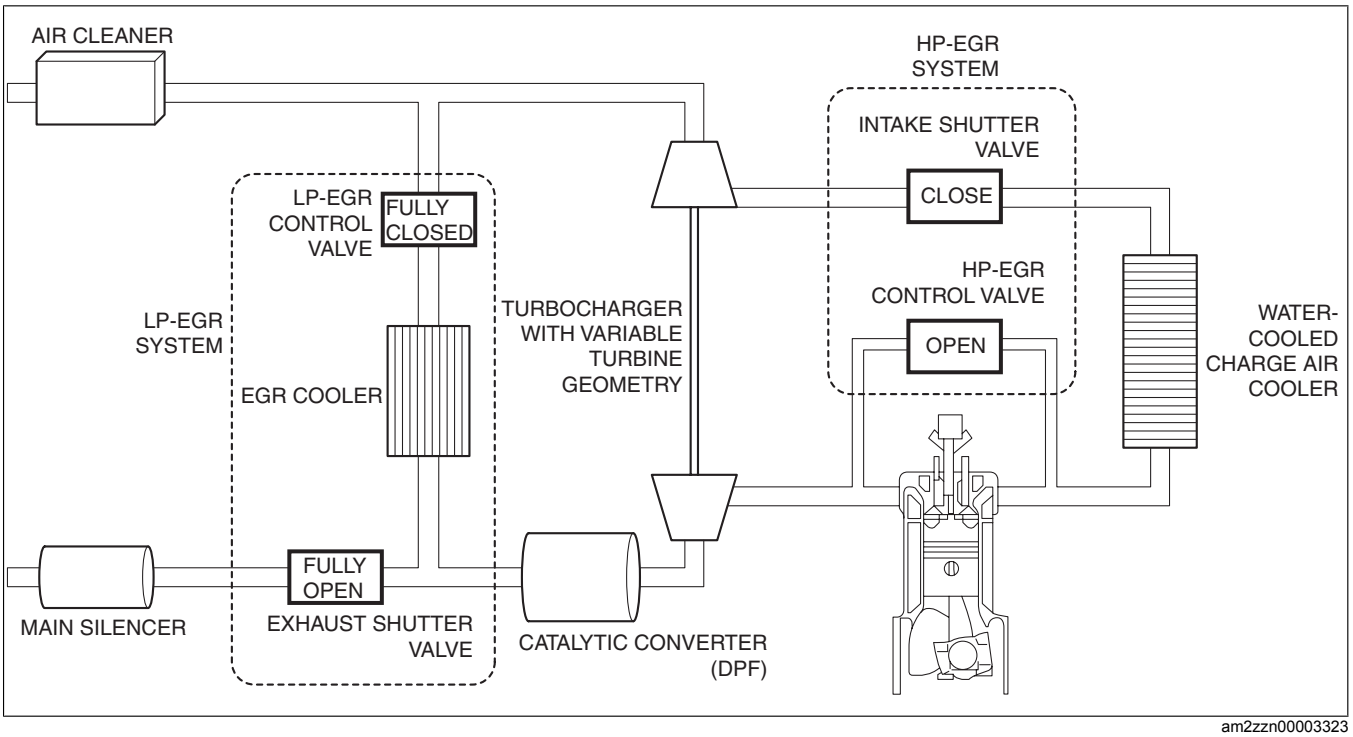
- The open/closed condition of each valve when the ignition is switched off is as follows.
  HP-EGR system
  — HP-EGR control valve: Fully closed
  — Intake shutter valve: Fully open
  LP-EGR system
  — LP-EGR control valve: Fully closed
  — Exhaust shutter valve: Fully open



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Light engine load

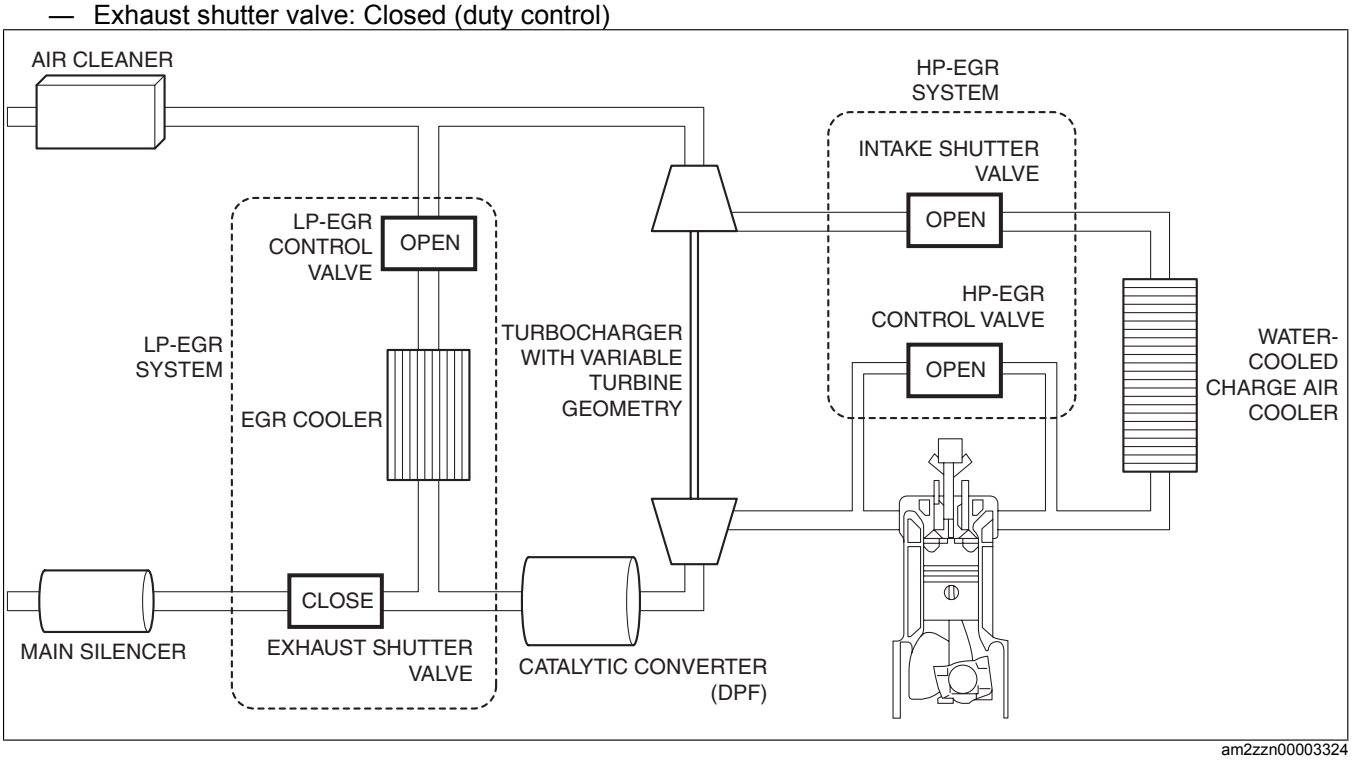
- During light engine loads (except during diesel particulate filter regeneration), exhaust gas is recirculated by the HP-EGR system according to the request flow amount from the PCM to promote engine warming and assure responsiveness.
- The open/closed condition of each valve during light engine loads is as follows.
  HP-EGR system
  — HP-EGR control valve: Open (duty control)
  — Intake shutter valve: Closed (duty control)
  LP-EGR system
  — LP-EGR control valve: Fully closed
  — Exhaust shutter valve: Fully open



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Medium engine load

- During medium engine loads (except during diesel particulate filter regeneration), exhaust gas is recirculated by both the LP-EGR and HP-EGR systems according to the request flow amount from the PCM.
- The open/closed condition of each valve during medium engine loads is as follows.
  HP-EGR system
  — HP-EGR control valve: Open (duty control)
  — Intake shutter valve: Open (duty control)
  LP-EGR system
  — LP-EGR control valve: Open (duty control)
  — Exhaust shutter valve: Closed (duty control)

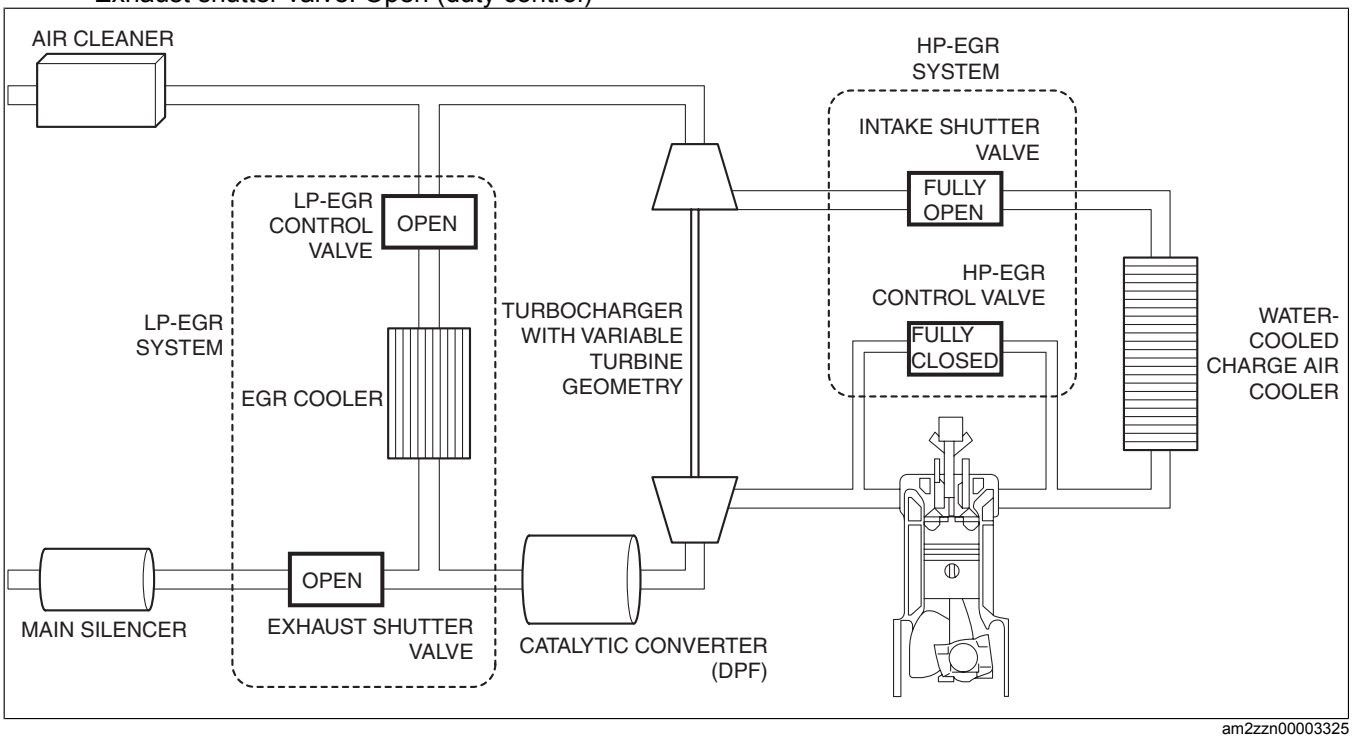


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High engine load

- During high engine loads (except during diesel particulate filter regeneration), the exhaust gas is recirculated by the LP-EGR system according to the request flow amount from the PCM to promote charging.

- The open/closed condition of each valve during high engine loads is as follows.
  HP-EGR system
  — HP-EGR control valve: Fully closed
  — Intake shutter valve: Fully open
  LP-EGR system
  — LP-EGR control valve: Open (duty control)
  — Exhaust shutter valve: Open (duty control)

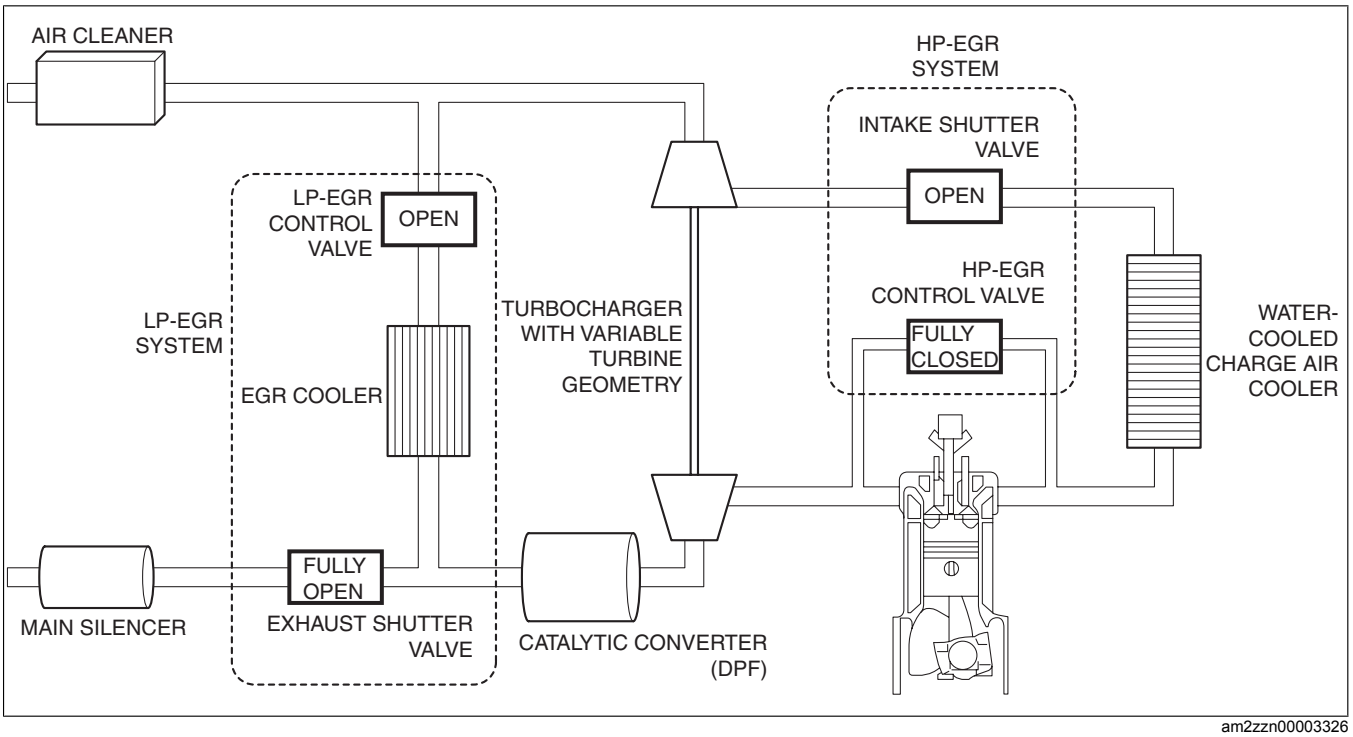


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During diesel particulate filter regeneration
- During diesel particulate filter regeneration, exhaust gas is recirculated by both the LP-EGR systems according to the request flow amount from the PCM.
- The open/closed condition of each valve during diesel particulate filter regeneration is as follows.
  HP-EGR system
  — HP-EGR control valve: Fully closed (constantly)
  — Intake shutter valve: Open (duty control) (during light engine load/medium engine load)*1
  LP-EGR system
  — LP-EGR control valve: Open (duty control) (during light engine load/medium engine load)*2
  — Exhaust shutter valve: Fully open (constantly)

*1 : During high engine loads, the intake shutter valve is fully open due to the large amount of air volume required.

*2 : Because the exhaust gas temperature is high during compulsory diesel particulate filter regeneration or under high engine load conditions, the LP-EGR control valve is fully closed to prevent combustion difficulties caused by the exhaust gas being recirculated.



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- For details on the exhaust gas recirculation system, refer to the CONTROL SYSTEM. (See EGR CONTROL [SKYACTIV-D 1.5].) (See INTAKE SHUTTER VALVE CONTROL [SKYACTIV-D 1.5].) (See EXHAUST SHUTTER VALVE CONTROL [SKYACTIV-D 1.5].)